AI-Based Stroke Phase Detection and Symmetry Analysis in Competitive Swimming Footage: A Proof of Concept

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Abstract

Computer vision analysis in aquatic environments presents unique challenges due to refraction, turbulence, and occlusion, which hinder the accuracy of standard motion tracking systems. This project proposes a non-intrusive AI-based proof of concept system for reliable pixel-to-meter scaling using markers in the background. The core contribution is a novel methodology for dynamic pixel-to-meter scaling using standard lane rope markers as calibration objects, addressing the limitations of fixed-camera setups in varying pool environments. This progress report outlines the development of the modular marker prototypes and the architectural design of the computer vision pipeline. It concludes by defining the evaluation protocols and the remaining milestones required to validate the system's robustness in real-world scenarios

Acknowledgements

Thanks

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1 Introduction

1.1 Problem Statement

Performance analysis in competitive swimming is traditionally reliant on manual observation or expensive, sensor-based systems that can be intrusive to the athlete [1] While computer vision offers a non-intrusive alternative, the aquatic environment introduces significant noise. Light refraction at the air-water interface, surface turbulence caused by the swimmer, and occlusions like bubbles and splashes degrade the quality of visual data [2, 3]. Furthermore, environment conditions, camera distances and angles vary significantly between venues, making consistent metric quantification, such as speed in meters/second, difficult without distinct calibration markers.

1.2 Motivation

There is a distinct lack of standardized, automated systems for extracting swimming metrics from video footage that do not require complex setups. A standard, simple, and well documented system capable of extracting metric data from video with minimal setup could act as a proof of concept, to prove the viability of such systems in aquatic environments. This project focuses on validating feasibility and robustness rather than achieving state-of-the-art performance.

1.3 Aims and Objectives

The primary aim of this project is to develop a Proof of Concept system that enables the extraction of swimming metrics from video footage. The specific objectives are:

1. To design, prototype, and create a modular, under-water marker design that maximizes underwater visibility.

2. To create and annotate a small dataset of swimming footage using the designed markers in the background.

3. To establish a pipeline for dynamic pixel-to-meter scale extraction using the markers for calibration.

4. To utilize the calibrated scale to provide useful kinematic metrics such as swimmer velocity and trajectory.

1.4 Project Structure

The remainder of this report is structured as follows.

Chapter 2– Background This chapter provides an overview of the core concepts and technological advancements relevant to the project.

Chapter 3– Literature Review This chapter reviews existing research and methodologies in the field of computer vision for sports analytics, with a focus on swimming performance analysis. It identifies gaps in current approaches and situates the proposed system within the broader context of the use of AI in sports technology.

Chapter 4– Methodology This chapter details the proposed methodology for developing the proof-of-concept system, including the design of underwater markers, dataset creation, and the calibration pipeline for dynamic scale extraction.

Chapter 5– Evaluation This chapter presents the evaluation of the proposed system, including assessments of each step of the pipeline, and an overall assessment of the system's performance in extracting swimming metrics from video footage.

Chapter 6– Conclusion and Future Work This chapter summarizes the findings of the project, discusses the implications of the results, and outlines potential directions for future research and development in this area.

2 Background

Existing literature in sports biomechanics highlights the importance of kinematic metrics like acceleration and hand trajectory. While computer vision research has successfully applied deep learning for detection, classification, segmentation, and human pose estimation (HPE), a critical limitation in many CV-based swimming systems is the conversion of distances from the pixel space to real-world units, often relying on relative metrics or requiring external sensors.

This chapter covers the relevant foundations upon which the proposed system is built, including object detection and classification, semantic segmentation, and human pose estimation. The chapter concludes with a brief summary of the discussed techniques.

2.1 Object Detection and Classification

Object detection is a fundamental computer vision task that involves identifying the presence and location of objects within an image. In the context of swimming analysis, this typically involves localizing the swimmer and specific environmental markers. Modern detectors have shifted from region-based convolutional neural networks (R-CNNs) to "one-stage" detectors to optimize for speed and real-time performance.

The YOLO (You Only Look Once) series represents the state-of-the-art in one-stage detection. The iterations used in this work, YOLOv8 and YOLO26 [4], introduces an anchor-free detection mechanism. Unlike previous versions that relied on predefined anchor boxes, anchor-free models predict the centre of an object directly, which reduces the complexity of the non-maximum suppression (NMS) stage. This results in improved performance on varying scales, which is a critical feature when tracking markers that change size due to camera distance or water refraction. In sports analytics, YOLO-based architectures are favoured for their ability to process high-frame-rate video without significant latency [5].

2.2 Semantic Segmentation

Semantic segmentation involves assigning a categorical label to every pixel in an image, facilitating a more granular understanding of the scene than bounding boxes. For aquatic environments, segmentation is used to isolate the swimmer's body from highly textured backgrounds or to precisely delineate markers amidst bubbles and surface turbulence.

A significant advancement in this field is the Segment Anything Model (SAM) [6]. SAM is a "foundation model" trained on over a billion masks, enabling zero-shot generalization to new domains, such as underwater footage, without additional training. This project utilizes SAM 2 [7], which extends this capability to the temporal domain through a memory-based attention mechanism. SAM 2 treats video as a continuous stream, using a “memory bank” to store features from previous frames. This allows the system to maintain a stable segmentation of markers even during transient occlusions or periods of significant water splash, addressing a core problem identified in the aquatic environment.

In addition to its role in tracking, SAM 2 was utilized in this research as a semi-automated annotation tool to facilitate dataset creation. Because manual pixel-level annotation of video sequences is prohibitively time-consuming, SAM 2’s temporal propagation feature was used to generate high-fidelity masks for underwater markers across several video sequences from a few manual prompts. These generated masks were then converted into bounding boxes and used to fine-tune a custom YOLO26 model for marker detection. This “model-in-the-loop” approach enables the creation of robust, domain-specific detectors with significantly reduced manual labelling effort. It bridges the gap between general-purpose foundation models and specialized sports analytics tasks.

2.3 Human Pose Estimation

Human Pose Estimation (HPE) is the process of identifying and tracking the coordinates of specific anatomical keypoints, such as joints and extremities. In swimming, HPE is used to calculate stroke frequency and joint angles. Traditionally, HPE utilized High-Resolution Nets (HRNet) to maintain high-resolution representations throughout the network.

However, recent research has transitioned toward Transformer-based architectures. ViTPose [8] utilizes a non-hierarchical Vision Transformer (ViT) [9] as its backbone. Unlike convolutional neural networks (CNNs) that process images through local receptive fields, Transformers use self-attention to capture long-range dependencies across the entire frame. This is particularly advantageous in swimming, where a swimmer’s limb may be extended far from the torso; ViTPose can leverage the global context of the swimmer’s body to accurately predict the position of a wrist or ankle that might be partially obscured by water refraction. This project utilizes the large-scale variant of ViTPose to ensure high precision in keypoint localization before converting these coordinates into metric data through marker-based linear scaling and baseline projection.

2.4 Motion Compensation and Ego-motion

In sports videography, unless the camera is rigidly fixed to a stationary mount, the resulting footage contains “ego-motion”—displacement caused by the movement of the camera itself. For swimming analysis, where a coach may follow the athlete along the poolside, this introduces a significant challenge: the pixel coordinates of the swimmer are a summation of the swimmer’s actual velocity and the camera’s displacement.

To resolve this, computer vision systems utilize stationary environmental features, such as lane ropes or pool-floor markings, as a global coordinate system. By tracking these fixed points using a homography matrix or simpler translation vectors [10], the system can perform background registration. This process effectively “stabilizes” the coordinate space, allowing for the extraction of the swimmer’s true kinematic trajectory relative to the pool rather than the image frame.

2.5 State Estimation and Trajectory Smoothing

Raw data extracted from deep learning models often contains high-frequency noise or “jitter” caused by partial occlusions, light refraction, or fluctuating confidence scores in the model’s output. To transform these noisy observations into a physically plausible trajectory, state estimation techniques are employed.

The Kalman Filter [11] is the standard recursive algorithm for this purpose. It operates in two phases: a “predict” step, which estimates the current state (position and velocity) based on previous data and Newtonian laws of motion, and an “update” step, which adjusts this prediction based on the new noisy measurement from the CV model. This approach is particularly effective in swimming because it can bridge gaps in detection—such as when a swimmer’s hand is completely submerged or hidden by a splash—by relying on the predicted momentum of the limb until it reappears.

2.6 Summary

The integration of YOLO for rapid detection, SAM 2 for robust temporal segmentation and automated dataset generation, and ViTPose for high-fidelity pose estimation forms a comprehensive, modular pipeline for swimming analysis. These tools collectively address the core challenges of the aquatic environment—namely visual noise, light refraction, and the historical lack of labeled underwater data—by leveraging recent breakthroughs in anchor-free detection, memory-based attention, and global spatial transformers.

3 Literature Review

3.1 Evolution of AI in Sports Analytics

Computer vision (CV) has fundamentally altered the landscape of sports analytics, transitioning from manual notation to automated, high-fidelity measurement. While early systems relied on background subtraction and handcrafted features, the current paradigm is dominated by Deep Learning (DL). As noted by [12], AI-driven systems are now the standard for extracting kinematic data and estimating pose in real-time.

However, a significant shift is currently underway, from task-specific CNNs to "Foundation Models", large-scale models trained on massive datasets that exhibit zero-shot generalization. This thesis utilizes models such as SAM 2 and ViTPose, which provide a level of geometric and spatial reasoning previously unavailable in sports-specific contexts.

3.2 An Adversarial Domain for Computer Vision

While field sports provide stable visual environments, the aquatic setting serves as an adversarial domain that stress-tests the limits of image processing. The primary challenge remains light refraction; the air-water interface creates non-linear geometric distortions that violate the fundamental pinhole camera model assumptions [13].

Beyond refraction, the environment is plagued by dynamic noise. As [3] highlights, measurements are strongly influenced by disruptive elements such as bubbles, splashes, and surface glare. These elements introduce severe, transient occlusions. Traditional trackers often fail in these high-turbulence zones. This necessitates the use of temporal memory mechanisms, such as the memory bank implemented in SAM 2 [7], which allows a system to remember the features of an object even when it is obscured by a splash or submerged beyond clear visibility.

3.3 From Handcrafted Features to Foundation Models

Historically, structural feature detection in swimming relied on the Hough Transform or colour-keying to isolate lane ropes or markers. However, these break down under varying pool illumination and water turbulence [14].

The subsequent shift toward Convolutional Neural Networks (CNNs), such as U-Net or earlier YOLO versions, offered better robustness to noise. For instance, [15] demonstrated the utility of YOLOv8 for athlete identification. Yet, CNNs are often limited by their local receptive fields, which can struggle with the long-range spatial dependencies required to identify a swimmer's full limb extension in a complex stroke.

This has led to the adoption of Vision Transformers (ViT). Models like ViTPose [8] utilize global self-attention mechanisms to process the entire frame simultaneously. This allows the model to leverage the context of the swimmer's torso to accurately predict the position of a wrist or ankle, even if that specific joint is visually distorted by surface ripples—a core technical choice reflected in this project's pipeline.

3.4 The Challenge of Metric Scaling and Calibration

A fundamental limitation in many CV-based swimming systems is the conversion of distance from pixel space to real-world, metric units. A velocity of x pixels per second is meaningless without a reference scale.

The gold standard remains 3D camera calibration, but the practicalities of poolside data collection often preclude this. Consequently, researchers have turned to planar homography using known environmental features. While [16] utilized lane ropes, recent work suggests that discrete, underwater markers provide a more stable reference. By establishing a fixed pixel-to-meter ratio based on known marker distances such as the 2.5m proposed in this thesis, systems can achieve metric accuracy without expensive sensor arrays. This project extends this logic by using SAM 2 to track these markers with high temporal consistency, ensuring the calibration scale remains valid across the entire video sequence.

3.5 Addressing the Data Scarcity Gap: Model-in-the-Loop Annotation

A persistent gap identified in the literature is the scarcity of publicly available, annotated underwater datasets. SwimmerNET [3] provides 2D keypoints but lacks underwater marker data. This creates a cold start problem for research.

A novel solution emerging in recent literature is Model-in-the-Loop (MITL) annotation. Rather than manual labelling, researchers use Foundation Models to "pseudo-label" datasets. By using SAM 2's zero-shot segmentation to identify markers in a few frames and propagating those masks through time, a high-quality training set can be generated for a more efficient detector like YOLOv8. This workflow, implemented in the current thesis, represents a departure from traditional manual annotation and addresses the research need for domain-specific underwater markers.

3.6 Summary of Research Needs

The reviewed literature confirms the following three persistent challenges. Traditional trackers lose the target during occlusions such as splashes, refraction renders simple linear scaling inaccurate, and that there is a lack of labelled data for underwater calibration features.

This research bridges these gaps by synthesizing SAM 2 for stable temporal segmentation, ViTPose for context-aware human pose estimation, and a custom YOLO26 detector trained via an automated MITL pipeline. By doing so, the project moves beyond qualitative video analysis toward a robust, metric-accurate proof-of-concept system for swimming biomechanical analysis.

4 Methodology

This chapter details the technical implementation of the proof-of-concept system. The methodology is structured around the four primary project objectives, transitioning from physical marker design to the final kinematic extraction engine.

4.1 Marker Design and Visibility

The first objective required a modular marker design capable of withstanding the high-turbulence environment of a swimming pool while remaining identifiable under varying refractive conditions.

The lack of existing documentation on underwater marker design for computer vision applications necessitated an iterative prototyping approach. This led to the decision to create a modular design for a 3d-printed marker that could be easily set up along the lane. The final design, as shown in Figure 4.1, consists of high-contrast, alternating black and white segments, which seamlessly screw into one another. This decision to use a modular design allowed us to test various configurations, mainly how many segments to stack, effectively altering the height of the marker.

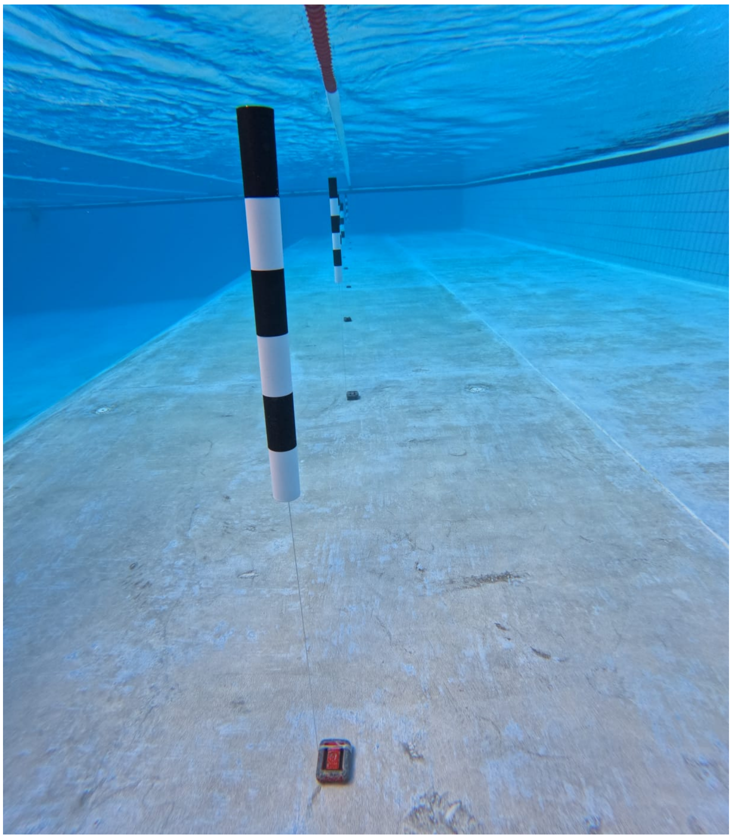

Figure 4.1 An image of the final marker design in use.

Each segment is 10 cm in height, with a 5 cm diameter, and the alternating pattern provides a unique visual signature that can be reliably detected by computer vision algorithms. The design also incorporates a screw mechanism that allows for easy assembly and disassembly, enabling quick setup along the lane for calibration purposes. This modularity not only facilitates testing different configurations but also ensures that the markers can be easily transported and stored when not in use.

The spacing between the markers was determined based on multiple tests, which revealed that an interval of 2.5m provided a good balance between visibility and the number of markers required for accurate calibration. This spacing allows for sufficient coverage along the lane while minimizing the number of markers needed, which is crucial for minimizing setup time and complexity.

4.2 Dataset Generation and Model-in-the-Loop Annotation

To address the second objective, necessitated by the lack of annotated underwater datasets, a "Model-in-the-Loop" (MITL) workflow was developed. This phase utilized foundation models to automate the labelling process for the downstream YOLO26 detector. The entire dataset creation pipeline is visualized in Figure 4.2.

4.2.1 Automated Labelling with SAM 2

The Segment Anything Model 2 (SAM 2) [7] was utilized to propagate marker bounding boxes across video sequences. Prompts were provided from the initial frames, propagated through the video. Once all tracked markers exited the frame, the process was repeated for the next chunk of frames until the entire video was annotated. This approach significantly reduced manual annotation time while maintaining high-quality labels. This process was repeated for multiple videos, resulting in a dataset of over 4,000 annotated frames, and over 7,000 bounding boxes.

4.2.2 Outlier Filtering

Automated tracking via foundation models, while efficient, is susceptible to "mask jitter" and drift, particularly when visual features are distorted by water refraction or transient occlusions. To ensure the robustness of the downstream YOLO detector, a statistical filtering stage was implemented to prune the raw SAM 2 output.

The filtering process focuses on the geometric consistency of the generated bounding boxes. Based on the assumption that the physical dimensions of the markers are constant, any significant variation in the predicted box area or aspect ratio was flagged as tracking noise. The system utilizes the Interquartile Range (IQR) method [17] to identify and remove these outliers. For a set of bounding box areas A, the filtering bounds are defined as:

$$[\text{Lower Bound}, \text{Upper Bound}] = [Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}] \tag{4.1}$$

where Q_1 and Q_3 represent the first and third quartiles, respectively. Any frame containing boxes falling outside these bounds, often caused by SAM 2 "bleeding" into a splash or expanding to cover the background, was excluded from the training set. This ensured that the model was trained only on high-confidence, geometrically plausible representations of the underwater markers.

4.2.3 Dataset Selection

To ensure the large YOLO26 model was trained using the best possible dataset, an A/B testing approach was implemented. Two datasets were created: Dataset A, which included all SAM-generated labels, and Dataset B, which included only the filtered labels after the IQR outlier removal process. Both datasets were used to train separate YOLO26n models, and their performance was compared using standard metrics such as mean Average Precision (mAP) and visual inspection of the predicted bounding boxes. The dataset that yielded the best performance was selected for the final training of the final YOLO26s model.

4.2.4 Human-in-the-Loop Verification

To ensure the high precision required for metric extraction, a custom-developed graphical verification tool was utilized to perform a final audit of the filtered dataset. This tool, implemented using the Tkinter framework, allowed for the rapid visual inspection of the SAM-generated labels.

The interface provided the functionality to manually adjust bounding box boundaries or delete erroneous detections that bypassed the statistical IQR filter. This hybrid approach of combining foundation model speed with human precision addressed the "domain-shift" problem inherent in aquatic footage, where surface glare can occasionally mirror markers. This step was crucial for establishing a ground truth used for the final model fine-tuning and the subsequent accuracy evaluations in Chapter 5.

4.3 Dynamic Scale Extraction Pipeline

Objective 3 focuses on establishing a stable reference for converting image coordinates into metric units.

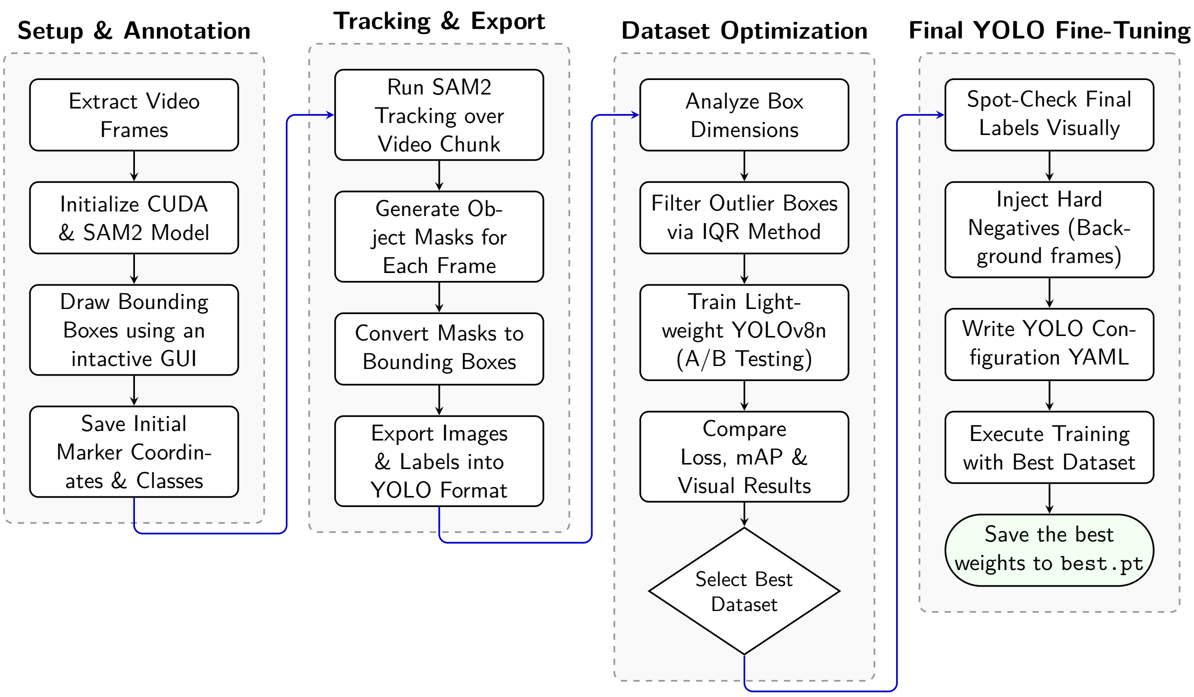

Figure 4.2 Marker Detection Model Training Pipeline.

4.3.1 Marker Detection via YOLO26

A custom YOLO26s model [4] was fine-tuned on the SAM-generated dataset to provide a robust method for localizing markers. To ensure high localization accuracy, which is critical for the subsequent pixel-to-meter ratio calculation, the training process utilized a Distribution Focal Loss (DFL) gain of 1.5 and a Box Loss gain of 7.5. These settings prioritize bounding box regression over classification accuracy, as the project involves a single-class detection task.

Training was conducted over 200 epochs with an image resolution of 640×640 pixels. To mitigate overfitting on the SAM-generated dataset, several regularization and augmentation strategies were employed:

- **Optimization Strategy:** An initial learning rate of 0.01 was used with a 3-epoch warm-up period and a momentum of 0.937 to stabilize early gradient descent.
- **Spatial Augmentations:** The model utilized Mosaic augmentation (1.0) and Random Erasing (0.4) to simulate occlusions and varying marker densities, forcing the network to learn robust features rather than memorizing specific marker placements.
- **Colour Distortions:** Variations in lighting and glare were accounted for by adjusting HSV (Hue, Saturation, Value) gains ($hsv\_h = 0.015, hsv\_s = 0.7, hsv\_v = 0.4$), ensuring the detector remains effective under inconsistent illumination.

- **Early Stopping:** A patience of 20 epochs was implemented to terminate training if the validation fitness failed to improve, ensuring the model generalizes well to real-world frames.

The resulting model outputs high-confidence bounding boxes that serve as the foundation for physical distance estimation, leveraging the known dimensions of the physical markers to establish spatial scale.

4.3.2 Refraction-Aware Scaling

As identified in preliminary experiments (Section 4.2), frame-independent scaling is susceptible to refractive noise. To stabilize the scale, the system implements a Median Absolute Deviation (MAD) inlier mask [18]. This filter calculates the pixel-to-meter ratio (px/m) for all visible marker pairs and rejects outliers that deviate significantly from the median, ensuring a stable calibration constant C for each frame.

4.4 Kinematic Metric Extraction

The final phase synthesises the outputs of marker detection, ego-motion compensation, and pose estimation to produce absolute positional and kinematic metrics for the swimmer.

4.4.1 Ego-Motion Compensation

Because the camera is hand-held and moved manually alongside the swimmer, raw pixel coordinates are subject to both lateral drift and incidental shake between frames, making them incomparable across time without compensation. To recover an absolute reference frame, the system exploits the lane markers as stationary anchors. For each consecutive frame pair, the horizontal pixel displacement of each detected marker is computed and matched to the nearest marker in the previous frame within a 50-pixel tolerance window. The median of all valid per-marker displacements gives a robust estimate of the camera's pixel shift Δx_{cam}, which is accumulated into a running global offset X_{cam}. The median is used in preference to the mean, providing robustness against the non-uniform jitter introduced by handheld camera movement:

$$X_{\text{cam}}^{(t)} = X_{\text{cam}}^{(t-1)} - \Delta x_{\text{cam}}^{(t)} \tag{4.2}$$

If no markers are detected in a given frame, the previous frame's displacement estimate is propagated forward (linear projection), preventing a discontinuity in the accumulated offset. The global pixel position of the swimmer is then:

$$p_{\text{global}} = p_{\text{pixel}} + X_{\text{cam}} \tag{4.3}$$

where p_{pixel} is the swimmer's raw horizontal pixel coordinate in the current frame.

4.4.2 Metric Scale Calibration

The pixel-to-metre conversion factor ρ (px/m) is estimated from the detected marker positions in each frame. Given N markers detected and sorted left-to-right, the outermost pair spans $N-1$ physical intervals of known separation d_{real} (metres). The scale is therefore:

$$\rho = \frac{\|\mathbf{p}_N - \mathbf{p}_1\|_2}{(N-1)\, d_{\text{real}}} \tag{4.4}$$

The estimate from the most recent frame in which two or more markers are visible is carried forward to frames where the marker count drops below two. The swimmer's absolute position in metres is then:

$$x_{\text{m}} = \frac{p_{\text{global}}}{\rho} \tag{4.5}$$

4.4.3 Swimmer Localisation via ViTPose

The swimmer's position is localised using the ViTPose pose estimation model [8]. Rather than using the bounding box centroid, the system extracts the left and right hip keypoints (COCO indices 11 and 12) and computes their midpoint as a proxy for the centre of mass (CoM):

$$\mathbf{c}_{\text{hip}} = \frac{\mathbf{k}_{11} + \mathbf{k}_{12}}{2} \tag{4.6}$$

This approach is more stable than bounding box tracking, as the hip midpoint is less affected by arm extension during the stroke cycle. To account for any lateral offset between the swimmer and the marker baseline, the hip centroid is orthogonally projected onto the line segment connecting the outermost detected markers before the position is converted to metres.

Wrist keypoints (COCO indices 9 and 10) are additionally extracted with a confidence threshold of 0.3, providing left- and right-wrist positions in the global reference frame. These are recorded alongside the hip CoM position for downstream stroke-cycle analysis.

4.4.4 Velocity and Kinematic Profile Extraction

The raw per-frame position records are first subjected to a Median Absolute Deviation (MAD) outlier filter and a temporal jump filter to remove implausible discontinuities. The cleaned trajectory is then processed by a discrete-time Kalman Filter [11], which models the swimmer as a constant-velocity system and simultaneously estimates position and instantaneous velocity from the noisy observations.

The instantaneous velocity is obtained from the Kalman state vector directly as the estimated rate of change of position. Equivalently, it may be expressed as the first derivative of the smoothed displacement with respect to time:

$$v(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{p_{\text{global}}(t)}{\rho}\right) \tag{4.7}$$

From the filtered state estimates, summary kinematic metrics are derived: absolute speed profile, peak velocity, and mean velocity over the analysed swim segment.

4.5 Summary

This chapter presented a four-phase methodology for extracting swimmer kinematics from handheld poolside video. First, a modular 3D-printed marker system was designed to remain visually identifiable under the refractive and turbulent conditions of the aquatic environment. Second, a Model-in-the-Loop annotation pipeline, combining SAM 2 propagation with IQR-based outlier filtering and human verification, was used to construct a domain-specific training dataset for a fine-tuned YOLO26s detector. Third, a refraction-aware MAD inlier filter was applied to stabilise the per-frame pixel-to-metre scale estimate derived from the known marker spacing. Finally, ego-motion compensation, ViTPose-based hip localisation, and a Kalman filter smoothing stage were combined to produce absolute positional and velocity profiles from the raw tracking output. Each phase directly addresses one of the project objectives and mitigates a specific source of error introduced by the aquatic recording environment.

5 Evaluation

5.1 Overview

This chapter evaluates the proof-of-concept system against quantitative ground truth data. Given the exploratory nature of the project, the evaluation focuses on a single representative swim sequence (participant P035, stroke breaststroke, pool length 25.0 m) in which the swimmer completes one full length and returns, yielding a 50 m trajectory. The primary metric is the temporal accuracy with which the system can recover the swimmer's position at known spatial checkpoints. Kinematic outputs, including velocity and wrist-relative motion profiles, are additionally presented and discussed.

5.2 Ground Truth Collection

Ground truth split times were collected retrospectively by reviewing the underwater video footage and manually recording the timestamp at which the swimmer's body crossed each 2.5 m marker boundary. This produced 10 outbound checkpoints (2.5 m to 25.0 m) and 10 return checkpoints (22.5 m back to 0.0 m), for a total of 20 reference observations. Times are expressed relative to the detected swim start, defined as the first frame in which sustained forward motion is observed. All splits were recorded by a single observer, thus no inter-rater reliability measure was applied, which represents an inherent limitation of the ground truth.

5.3 Trajectory Optimisation and Pre-Evaluation Processing

5.3.1 Outlier Removal

Prior to evaluation, the raw tracking records were subjected to the MAD-based inlier filter and temporal jump filter described in Chapter 4. Of the 4,052 raw records, 723 were flagged as significant deviations and removed, leaving 3329 records for downstream processing. A secondary velocity ceiling of 5.0 m/s was subsequently applied, removing a further 32 records whose implied speed was physiologically implausible for competitive breaststroke swimming.

5.3.2 Loop Closure Correction

The system detects the turn point as the frame index at which the Kalman-filtered position estimate reaches its maximum. For this sequence, the turn was detected at $t = 44.92$

s with a raw maximum position of 27.12 m, requiring an outbound scale correction factor of 0.9218 to anchor the peak exactly at 25.0 m. The return trajectory was subsequently re-mapped using a linear interpolation between the corrected peak and the expected return endpoint. Following loop closure, a Savitzky–Golay filter (window length 61, polynomial order 3) was applied to the centroid velocity signal to suppress residual high-frequency noise.

It should be noted that the two endpoint splits, 25.0 m on the outbound leg and 0.0 m on the return, could not be recovered and are reported as N/A in Table 5.1. The threshold-crossing detection searches for the first frame at which the trajectory strictly exceeds a target distance; because the trajectory is noisy near the turn and near the pool wall, it does not cleanly cross these boundary values, and no valid crossing index is found.

5.4 Positional Accuracy Evaluation

5.4.1 Split-Time Comparison

Table 5.1 presents the per-checkpoint comparison between the system's automatically derived split times and the manually recorded ground truth. The error is defined as:

$$e_i = t_{\text{auto},i} - t_{\text{true},i} \tag{5.1}$$

where a negative value indicates that the system reached checkpoint i earlier than the ground truth observation.

5.4.2 Analysis of Errors

The system achieves an overall MAE of 1.005 s and an RMSE of 1.210 s across 18 recoverable checkpoints. The error distribution exhibits distinct fluctuations rather than a uniform trend across phases. On the outbound leg, the system initially overestimates swimming speed in the mid-pool section—arriving at checkpoints early and reaching a peak error of -2.70 s at the 15.0 m mark. However, this trend reverses towards the end of the lap, where the system underestimates speed and arrives late, generating a $+2.02$ s error at 22.5 m. This suggests non-linear tracking distortions or fluctuating pixel-to-metre scale estimations rather than a simple accumulating drift. On the return leg, errors are slightly smaller in magnitude and more consistent; after initial late arrivals near the turn, the system reliably arrives slightly early, with errors clustering between -0.58 s and -1.48 s. The full trajectory comparison against ground truth checkpoints is shown in Figure 5.1.

Table 5.1 Updated per-checkpoint split time comparison: automated system vs. ground truth. Error $= t_{\text{auto}} - t_{\text{true}}$. (Start at index 385, Turn at index 2041)

Phase	Distance (m)	True Time (s)	Auto Time (s)	Error (s)
Outbound	2.5	2.00	2.12	$+0.12$
Outbound	5.0	4.00	3.88	-0.12
Outbound	7.5	7.00	5.37	-1.63
Outbound	10.0	9.00	8.75	-0.25
Outbound	12.5	12.00	10.15	-1.85
Outbound	15.0	14.00	11.30	-2.70
Outbound	17.5	17.00	16.03	-0.97
Outbound	20.0	20.00	20.50	$+0.50$
Outbound	22.5	22.00	24.02	$+2.02$
Outbound	25.0	26.00	N/A	N/A
Return	22.5	27.00	28.13	$+1.13$
Return	20.0	29.00	29.90	$+0.90$
Return	17.5	32.00	31.37	-0.63
Return	15.0	35.00	33.52	-1.48
Return	12.5	38.00	37.17	-0.83
Return	10.0	41.00	40.07	-0.93
Return	7.5	43.00	42.20	-0.80
Return	5.0	46.00	45.42	-0.58
Return	2.5	49.00	48.37	-0.63
Return	0.0	51.00	N/A	N/A
Overall MAE				**1.005 s**
Overall RMSE				**1.210 s**

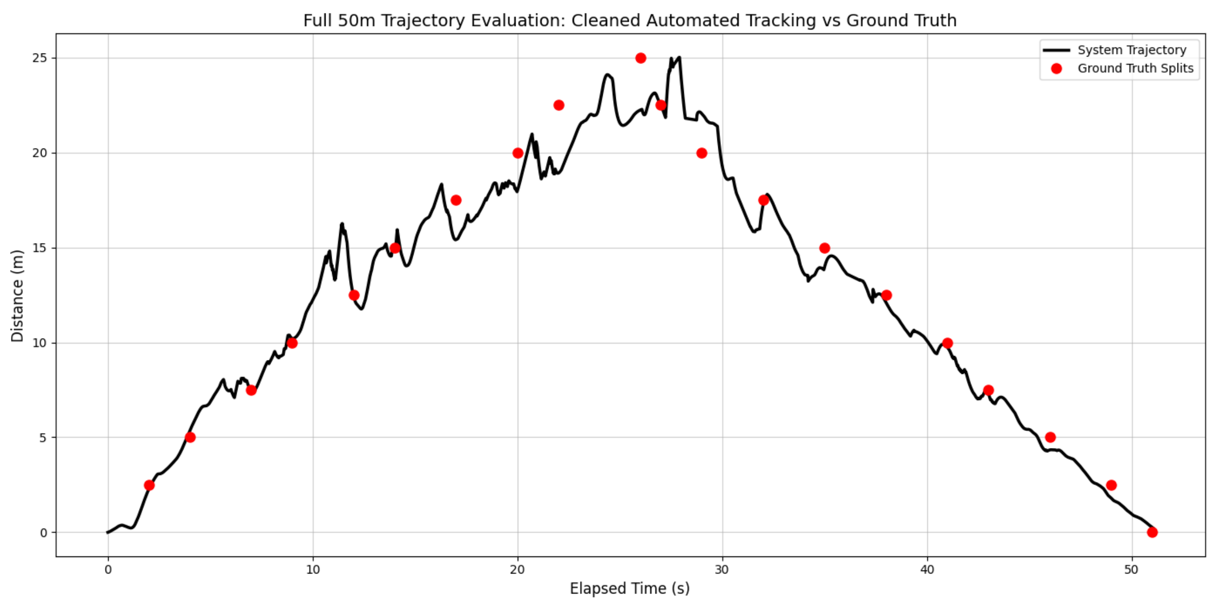

Figure 5.1 Full 50 m trajectory evaluation: system trajectory (black line) versus manually recorded ground truth split times (red points). The swimmer travels from 0 m to 25 m (outbound) and returns to 0 m.

5.5 Kalman Filter Noise Reduction

As an internal validation of the filtering pipeline, Table 5.2 reports the standard deviation of the raw and Kalman-smoothed estimates for both position and the pixel-to-metre scale factor. The Kalman filter achieves a modest reduction in position variance and a more substantial reduction in scale variance, reflecting its effectiveness at rejecting the frame-to-frame refractive noise that affects the marker-based scale estimate.

Table 5.2 Raw vs. Kalman-filtered standard deviation for position and scale estimates.

Signal	Raw σ	Kalman σ
Position (m)	6.732	6.706
Scale (px/m)	33.364	26.669

5.6 Kinematic Outputs

5.6.1 Centroid Velocity Profile

Following loop closure and smoothing, the swimmer's peak speed was estimated at 6.57 m/s (23.65 km/h) and the mean speed at 1.59 m/s (5.73 km/h). These are derived from the Savitzky–Golay smoothed velocity signal rather than the raw Kalman output; the pre-closure Kalman estimate yielded a higher peak of 8.44 m/s, which is physiologically implausible for competitive swimming and reflects residual noise prior to the final smoothing stage. The corrected figures are more consistent with reported competitive breaststroke velocities for recreational to trained swimmers. The full velocity and speed profiles are shown in Figure 5.2.

5.6.2 Wrist Kinematics and Symmetry

To isolate the intra-stroke cycle mechanics from the swimmer's macroscopic forward translation, wrist kinematics were evaluated by subtracting the centroid position from the raw wrist coordinates. This centroid-anchored view is intended to reveal the cyclical pull and recovery phases relative to the swimmer's body. However, as visible in Figure 5.2, the resulting relative wrist signals exhibit substantial high-frequency noise and spike artefacts throughout the sequence, persisting despite biological thresholding, median filtering, and Savitzky–Golay smoothing. This instability highlights the limitations of the ViTPose keypoint predictions in challenging underwater environments, where partial occlusion, splash, and motion blur frequently cause distal keypoints like the wrists to be dropped or mislocated between frames.

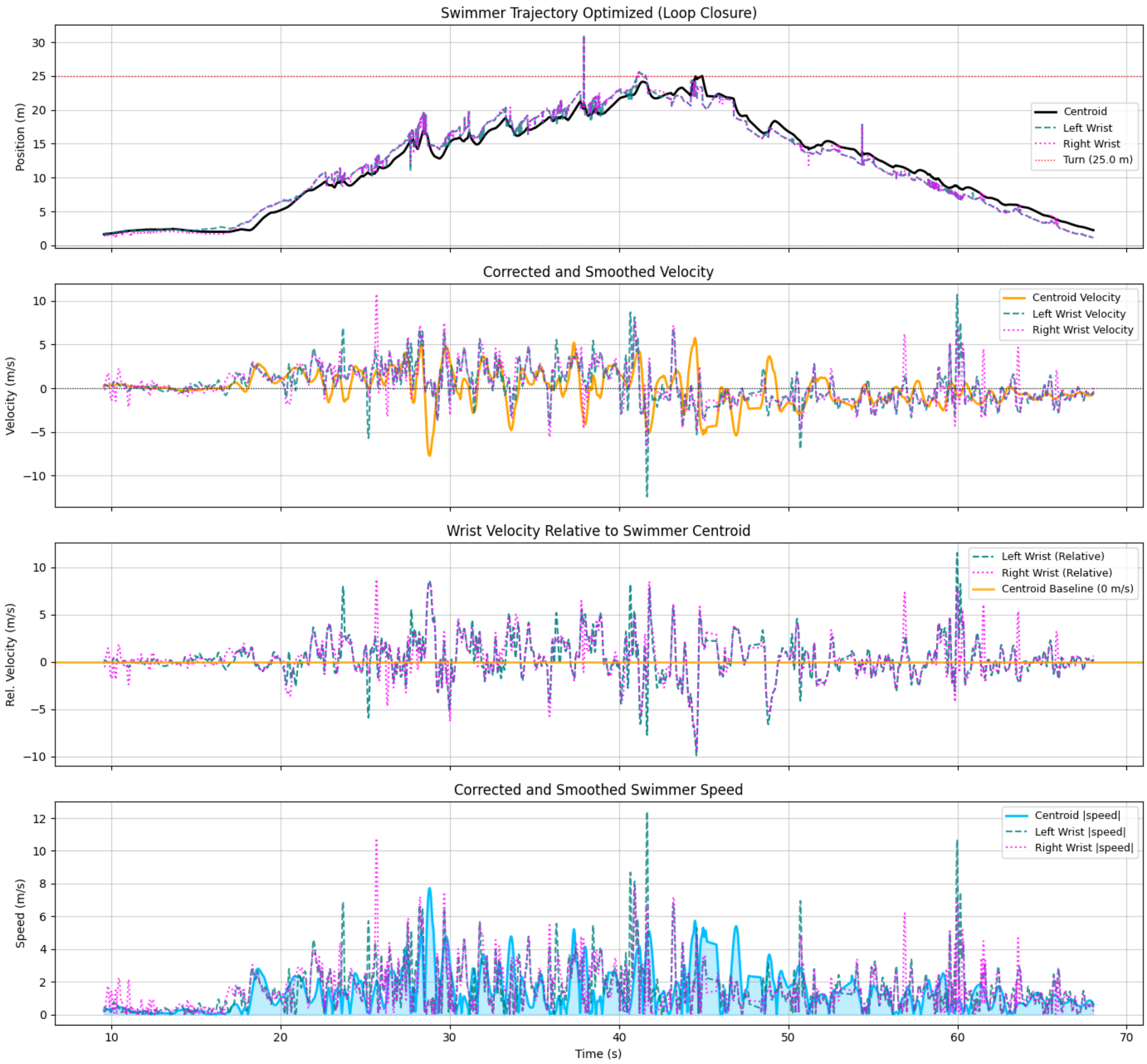

Figure 5.2 Kinematic profiles for the full 50 m sequence: (top) loop-closure-corrected trajectory for centroid and both wrists; (second) smoothed velocity; (third) wrist velocity relative to swimmer centroid; (bottom) absolute speed magnitude.

Despite the inherent noise in individual trajectories, evaluating the coordination between limbs provides valuable insights. Because breaststroke dictates simultaneous and symmetrical arm movements, the relationship between the left and right wrists was visualised using a symmetry heatmap (Figure 5.3). This heatmap plots the spatial correspondence between the two limbs over the duration of the swim. In an ideal stroke with perfect tracking, the distribution would manifest as a highly concentrated, mirrored density.

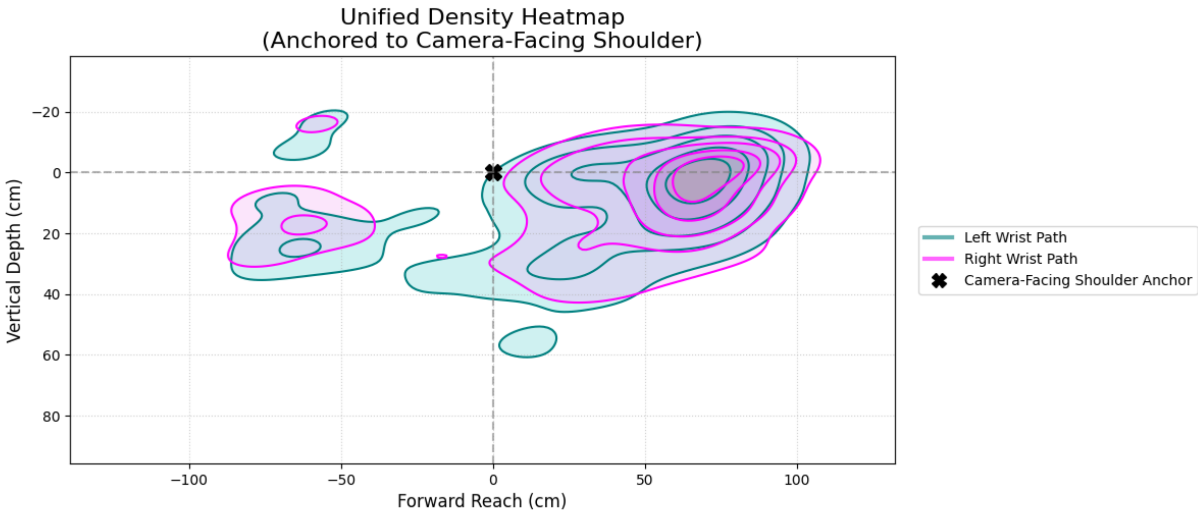

Figure 5.3 Symmetry heatmap illustrating the spatial correlation between the left and right wrist motions relative to the swimmer's centroid. High-density areas indicate the most common spatial relationship between the limbs during the stroke cycle.

The generated heatmap exhibits distinct central densities that indicate the underlying stroke rhythm and the overall synchronous nature of the breaststroke pull. However, the presence of asymmetrical dispersion and outliers in the heatmap further corroborates the tracking noise observed in the velocity profiles. While the macro-level phase correspondence between the arms is correctly captured by the model—remaining roughly in phase and appropriately positioned in front of the waist—the fine-grain kinematic noise limits the utility of the current system for precision technique analysis.

5.7 Summary

The system achieves a mean absolute split-time error of 1.005 s and an RMSE of 1.210 s across 18 checkpoints over a 50 m breaststroke sequence, demonstrating that marker-based ego-motion compensation and Kalman filtering can produce a usable positional trajectory from handheld, uncalibrated poolside footage. The primary positional limitation identified is the non-linear fluctuation in scale estimation on the outbound leg—resulting in alternating periods of overestimated and underestimated swimming speed—rather than a strict, continuous monotonic drift. Distal keypoint tracking, particularly for the wrists, remains noisy under water and represents the primary target for future refinement if detailed, centroid-relative technique analysis is desired.

6 Conclusion

6.1 Project Summary

This project successfully developed an AI-based proof-of-concept system for the non-intrusive extraction of swimming kinematics from uncalibrated, handheld poolside video footage. Driven by the need to overcome severe visual distortions present in aquatic environments, such as refraction, turbulence, and occlusion, a modular computer vision pipeline was proposed. By introducing a novel dynamic pixel-to-meter scaling methodology utilizing custom-designed underwater markers, the system demonstrated the feasibility of extracting real-world metric data without relying on rigid camera setups. The integration of advanced deep learning foundation models, including SAM 2 for temporal segmentation, YOLO26 for marker detection, and ViTPose for human pose estimation, provided a robust framework for tracking swimmer movements and addressing domain-specific data scarcity.

6.2 Achievement of Objectives

The project successfully met its core objectives:

1. **Marker Design:** A highly visible, modular 3D-printed marker system was developed to withstand underwater turbulence and provide a reliable calibration reference at 2.5 m intervals.
2. **Dataset Generation (MITL):** A Model-in-the-Loop annotation pipeline leveraging SAM 2, statistical outlier filtering (IQR), and human verification was implemented to construct a domain-specific dataset for training a custom YOLO26s detector.
3. **Dynamic Scale Extraction:** A refraction-aware pipeline was established, utilizing Median Absolute Deviation (MAD) filtering to stabilize the pixel-to-meter scale derived from the known marker spacing.
4. **Kinematic Metric Extraction:** Combining ego-motion compensation, ViTPose localization, and a Kalman filter smoothing stage, the system successfully produced absolute positional and velocity profiles from the raw tracking output.

6.3 Limitations and Challenges

Despite proving the feasibility of the concept, the evaluation highlighted several inherent challenges in the aquatic domain. The positional accuracy evaluation on a 50 m breaststroke sequence yielded a Mean Absolute Error (MAE) of 1.005 s and a Root Mean Square Error (RMSE) of 1.210 s across 18 checkpoints. A primary limitation was the non-linear fluctuation in scale estimation on the outbound leg, leading to alternating periods of overestimated and underestimated swimming speeds. Furthermore, while macroscopic centroid tracking was successfully extracted, tracking distal keypoints such as the wrists remained noisy underwater and exhibited substantial high-frequency artefacts. Consequently, fine-grain kinematic noise limited the utility of the current system for precision technique analysis. Lastly, the evaluation relied on single-observer manual ground truth split times, which inherently introduces a limitation regarding inter-rater reliability.

6.4 Future Work

Building upon this proof of concept, several avenues for future research and refinement have been identified:

- **Enhancement of Pose Estimation:** Future iterations should target the refinement of distal keypoint tracking (particularly for wrists) to enable detailed, centroid-relative technique analysis.
- **Advanced Scale Stabilisation:** Methods to mitigate the non-linear scaling fluctuations caused by severe refraction should be explored to improve the continuous monotonic accuracy of the estimated swimming speed.
- **Broader Dataset Validation:** While the proof of concept successfully evaluated a breaststroke sequence, future work should validate the system's robustness across different strokes, variable lighting, and different pools.
- **Automated Ground Truth Collection:** Replacing manual, single-observer timestamp recording with sensor-based timing systems (such as touchpads) would provide a more objective baseline for positional accuracy evaluation.

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Appendix A Sample A

Appendix B Sample B